PH 101

Lecture 09

OUTLINE

- Generalized coordinates and generalized velocities
- The Principle of Least action
- Lagrange equations of motion

Generalized coordinates and Generalized velocities

Any n quantities q_1, q_2 , ... q_n which completely define the position of a system with n degrees of freedom are called generalized coordinates of the system, and the derivatives $\dot{q}$ are called its generalized velocities.

It is convenient to regard the list of values q_1, q_2 , ... q_n as the coordinates of a 'point' **q** in a space of n dimensions, that is:

$$\boldsymbol{q} = (q_1, \ldots, q_n)$$

Such a space is termed as configuration space, denoted by the symbol $\mathcal{Q}$

Let S be the two-particle system shown in Fig. 1. This system has two degrees of freedom and generalized coordinates x, θ. In this case the configuration space Q is the (x, θ)-plane. Each point q = (x, θ) lying in Q corresponds to a configuration of the mechanical system S. Moreover, as the configuration of the system changes with time, the point q moves through the configuration space as shown in Fig. 2.

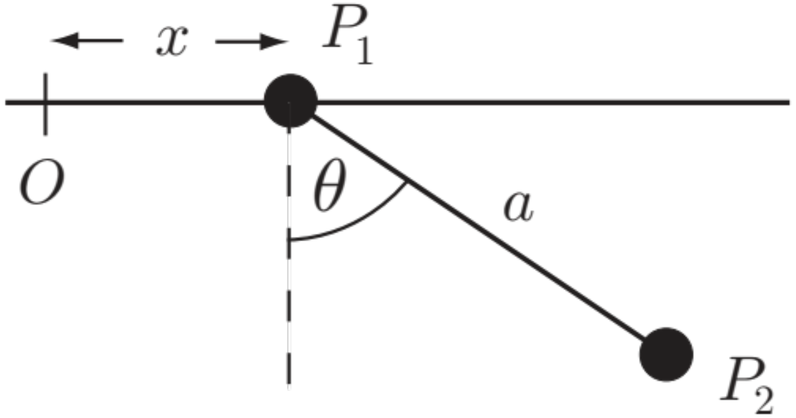

Fig. 1

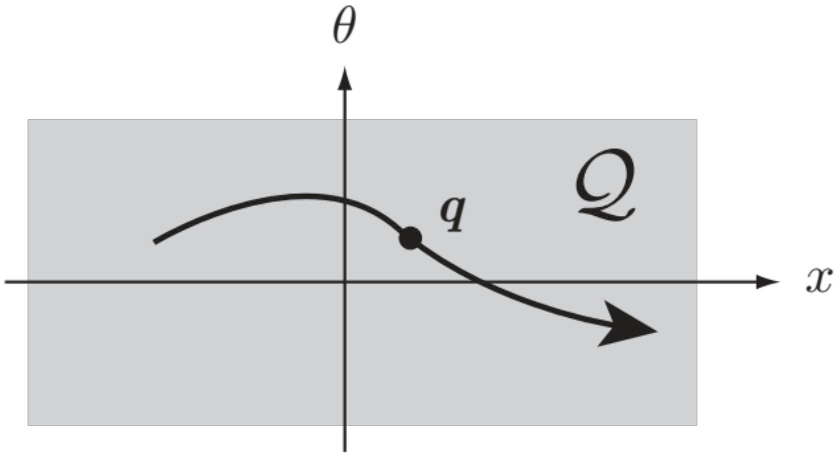

Fig. 2

The time derivatives $\dot{q}_1, \dot{q}_2 \ldots, \dot{q}_n$ of the generalized coordinates q_1, q_2 , ... q_n are called the generalized velocities of the system S.

The n-dimensional vector $(\dot{q}_1, \dot{q}_2 \ldots, \dot{q}_n)$, formed from the $\{\dot{q}_j\}$, is just the time derivative of the vector $\boldsymbol{q}$, that is,

$$\dot{\boldsymbol{q}} = (\dot{q}_1, \ldots, \dot{q}_n)$$

The vector $\dot{\boldsymbol{q}}$ can be regarded as the 'velocity' of the point **q** as it moves through the configuration space $\mathcal{Q}$.

- By the generalized coordinates q_1, q_2 , ... q_n we shall mean a set of independent coordinates, equal in number to the n degrees of freedom of the system.
- We use the word "generalized" to emphasize the fact that such coordinates are not necessarily of the type of the simple (x, y, z) or (r, θ, ϕ) systems and to indicate that they are not necessarily lengths or angles, but may be any quantity appropriate to the description of the position of the system.

The coordinates (x_k, y_k, z_k) of a point k are expressible in terms of the generalized coordinates $(q_1, q_2$, ... $q_n)$ by functional relations:

$$x_k = x_k(q_1, q_2 , ... q_n),$$
$$y_k = y_k(q_1, q_2 , ... q_n),$$
$$z_k = z_k(q_1, q_2 , ... q_n),$$

Or,

$$\vec{r}_k = \vec{r}_k(q_1, q_2, \dots q_n)$$

In general, when time is also involved, we write: $\vec{r}_k = \vec{r}_k(q_1, q_2, \dots q_n, t)$

To illustrate by an example:

If (q_1, q_2, q_3) are the cylindrical coordinates of a point $(r, \theta, \overset{z}{\cancel{\varphi}})$,

the foregoing equations become:

$$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z.$$

The Principle of Least Action

(Also known as Hamilton's principle)

Every mechanical system is characterized by a definite function
$L\ (q_1, q_2\ , \ldots .\ q_n; \dot{q}_1, \dot{q}_2 \ldots, \dot{q}_n, t)$ or briefly $L(q, \dot{q}, t)$
and the motion of the system is such that a certain condition is satisfied.

Let the system occupy, at the instants t_1 and t_2 positions defined by two sets of values of the co-ordinates, $q^{(1)}$ and $q^{(2)}$. Then the condition is that the system moves between these positions in such a way that the integral

$$S = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, \mathrm{d}t \qquad \textbf{(1)}$$

takes the least possible value The function L is called the Lagrangian of the system concerned, and the integral (1) is called the **Action**.

The fact that the Lagrangian contains only q and $\dot{q}$, but not the higher derivatives, expresses the result already mentioned, that **the mechanical state of the system is completely defined when the co-ordinates and velocities are given.**

Let us now derive the differential equations which solve the problem of minimizing the integral (1). For simplicity, we shall at first assume that the system has only one degree of freedom, so that only one function **q(t)** has to be determined.

Let q = q(t) be the function for which S is a minimum. This means that S is increased when q(t) is replaced by any function of the form

$$q(t) + \delta q(t) \qquad (2)$$

where $\delta q(t)$ is a function which is small everywhere in the interval of time from t_1 to t_2; $\delta q(t)$ is called a variation of the function **q(t)**. Since, for $t = t_1$ and for $t = t_2$, all the functions (2) must take the values $q^{(1)}$ and $q^{(2)}$ respectively, it follows that:

$$\delta q(t_1) = \delta q(t_2) = 0 \qquad (3)$$

The change in **S** when q is replaced by $q(t) + \delta q(t)$ is

$$\int_{t_1}^{t_2} L(q+\delta q, \dot{q}+\delta\dot{q}, t)\,\mathrm{d}t - \int_{t_1}^{t_2} L(q, \dot{q}, t)\,\mathrm{d}t$$

When this difference is expanded in powers of δq and $\delta\dot{q}$ in the integrand, the leading terms are of the first order. The necessary condition for S to have a minimum is that these terms (called the first variation, or simply the variation, of the integral) should be zero. Thus the principle of least action may be written in the form:

$$\delta S = \delta \int_{t_1}^{t_2} L(q, \dot{q}, t)\,\mathrm{d}t = 0, \qquad (4)$$

or, effecting the variation,

$$\int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q}\delta q + \frac{\partial L}{\partial \dot{q}}\delta\dot{q}\right)\mathrm{d}t = 0.$$

$$\int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right) \mathrm{d}t = 0.$$

Since $\delta \dot{q} = \frac{d\delta q}{dt}$, we obtain, on integrating the second term by parts,

$$\delta S = \left[\frac{\partial L}{\partial \dot{q}} \delta q \right]_{t_1}^{t_2} + \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t} \frac{\partial L}{\partial \dot{q}} \right) \delta q \, \mathrm{d}t = 0. \tag{5}$$

The conditions (3) show that the integrated term in (5) is zero. There remains an integral which must vanish for all values of δq. This can be so only if the integrand is zero identically. Thus we have:

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0}$$

When the system has more than one degree of freedom, the n different functions $q_i(t)$ must be varied independently in the principle of least action. We then evidently obtains equations of the form:

$$\boxed{\frac{\mathrm{d}}{\mathrm{dt}}\left(\frac{\partial L}{\partial \dot{q}_i}\right)-\frac{\partial L}{\partial q_i}=0} \qquad (i=1,2,\ldots,n) \qquad \textbf{(6)}$$

These are the required differential equations, called in mechanics Lagrange's equation

If the Lagrangian of a given mechanical system is known, the equations (6) give the relations between accelerations, velocities and coordinates, i.e. they are the equations of motion of the system.

Next class, we will discuss many-many examples to understand the concepts discussed in this class.

The variation in S and the meaning of 'stationary'

The function $q_s(t)$ is said to make the functional, $S[q]$ stationary if

$$S[q_s + \delta q] - S[q_s] = O(||\delta q||^2)$$

when $||\delta q||$ is small.

Example:

A certain oscillator with generalised coordinate q has Lagrangian

$$L = \tfrac{1}{2}\dot{q}^2 - \tfrac{1}{2}q^2.$$

Verify that $q_s = \sin t$ is a motion of the oscillator, and show directly that it makes the action functional $S[q]$ stationary in any time interval $[0, \tau]$.

Solution:

$$L = \frac{1}{2}\dot{q}^2 - \frac{1}{2}q^2$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = 0$$

$$\Rightarrow \quad \ddot{q} + q = 0$$

This is the classical SHM equation

Clearly $q = \sin t$ is a possible motion of the system. //

The action S for the time interval $[0, \tau]$ is

$$S[q] = \int_0^\tau \left(\frac{1}{2}\dot{q}^2 - \frac{1}{2}q^2\right) dt$$

$$S[q] = \int_0^{\tau} \left(\frac{1}{2}\dot{q}^2 - \frac{1}{2}q^2\right) dt$$

By the principle of least action the motion $q_s = \sin t$ makes S stationary. Let's prove it from the first principle.

consider the function

$$q = q_s + \delta q \quad , \text{ where } \delta q \text{ is}$$

a small variation such that $\delta q(0) = 0 = \delta q(\tau)$.

Then,

$$S[q_s + \delta q] = \frac{1}{2}\int_0^\tau \left[(\dot{q}_s + \delta \dot{q})^2 - (q_s + \delta q)^2 \right] dt$$

$$= \frac{1}{2}\int_0^\tau \left[(\cos t + \delta \dot{q})^2 - (\sin t + \delta q)^2 \right] dt$$

$$S[q_s+\delta q] = \frac{1}{2}\int_0^{\tau}\Big[\cos^2 t + \delta\dot{q}^2 + 2\cos t\ \delta\dot{q} - \sin^2 t - \delta q^2 - 2\sin t\ \delta q\Big]dt$$

Now,

$$S[q_s+\delta q] - S[q_s]$$

$$= \frac{1}{2}\int_0^{\tau}\Big[\cos^2 t + \delta\dot{q}^2 + 2\cos t\ \delta\dot{q} - \delta q^2 - 2\sin t\ \delta q - \sin^2 t - \cos^2 t + \sin^2 t\Big]dt$$

$$= \frac{1}{2}\int_0^{\tau}\left(2\delta\dot{q}\cos t + \delta\dot{q}^2 - 2\delta q\sin t - \delta q^2\right)dt$$

$$= \Big[\delta q\cos t\Big]\Big|_0^{\tau} + \frac{1}{2}\int_0^{\tau}\left(\delta\dot{q}^2 - \delta q^2\right)dt$$

Since $\quad \delta q(\tau) = 0 = \delta q(0)$

we get :

$$S[q_s + \delta q] - S[q_s] = \frac{1}{2}\int_0^\tau \left(\delta\dot{q}^2 - \delta q^2\right) dt$$

$$\equiv \underbrace{O\left(\|\delta q\|^2\right)}$$

on the order of $\|\delta q\|^2$

Thus in accordance with the principle of least action, the motion $q_s = \sin t$ makes the Action functional stationary.